

# Synthetic Historical Well Drilling Report

SYNTHETIC DEVELOPMENT DOCUMENT. This report is generated from the NWIS MVP synthetic database for demonstration and testing. It is NOT a real Oil India Limited drilling report and must not be treated as operational or confidential OIL material.

## Well identification

Well ID: W004. Field: Field-A. Well type: Development. Status: Completed. Total depth: 4490.0 m. Surface location (synthetic): latitude 23.656096, longitude 86.860862.

## Formation intervals

Formation-A: 0.0-1106.1 m. Lithology: Siltstone. Pressure: 36.41. Temperature: 45.57.

Formation-B: 1110.0-2226.6 m. Lithology: Sandstone. Pressure: 50.34. Temperature: 64.47.

Formation-C: 2225.7-3368.0 m. Lithology: Shale. Pressure: 57.93. Temperature: 77.59.

Formation-D: 3354.5-4490.0 m. Lithology: Limestone. Pressure: 67.41. Temperature: 99.85.

## Drilling observations

This well has 150 synthetic drilling-log samples in the NWIS database. The following observations summarize those records by formation interval. No additional drilling events are inferred beyond the stored logs and events table.

Drilling observations in Formation-A (0.0-1106.1 m, lithology Siltstone): 23 synthetic log samples. Mean ROP 31.37 m/h, mean WOB 10.34, mean RPM 115.33, mean torque 15.24, mean standpipe pressure 2388.97, mean mud density 1.09. Non-normal event\_label rows in this interval: 0.

Drilling observations in Formation-B (1110.0-2226.6 m, lithology Sandstone): 42 synthetic log samples. Mean ROP 25.54 m/h, mean WOB 12.19, mean RPM 114.26, mean torque 17.00, mean standpipe pressure 2510.70, mean mud density 1.11. Non-normal event\_label rows in this interval: 7.

Drilling observations in Formation-C (2225.7-3368.0 m, lithology Shale): 43 synthetic log samples. Mean ROP 25.42 m/h, mean WOB 12.36, mean RPM 114.96, mean torque 16.86, mean standpipe pressure 2574.56, mean mud density 1.14. Non-normal event\_label rows in this interval: 0.

Drilling observations in Formation-D (3354.5-4490.0 m, lithology Limestone): 43 synthetic log samples. Mean ROP 19.83 m/h, mean WOB 14.19, mean RPM 112.92, mean torque 19.66, mean standpipe pressure 2732.66, mean mud density 1.16. Non-normal event\_label rows in this interval: 7.

## Historical events

2 historical event record(s) are stored for W004. Each item below is copied from the synthetic events table.

Historical event at 1544.4 m in Formation-B. Event type: HIGH\_TORQUE. Severity: MEDIUM. Observation: Torque increased significantly while drilling through the interval.. Cause: Increased formation resistance and poor hole-cleaning conditions.. Mitigation: Adjusted WOB/RPM and improved circulation parameters.. Outcome: Torque returned toward the normal operating range..

Historical event at 4222.2 m in Formation-D. Event type: STUCK\_PIPE. Severity: HIGH. Observation: Drill string experienced elevated resistance and became temporarily stuck.. Cause: Increasing torque and declining rate of penetration in a mechanically unstable interval.. Mitigation: Circulation, controlled WOB reduction, and reciprocation of the drill string.. Outcome: Drilling

resumed after controlled freeing operation..

### **Source statement**

All well, formation, drilling-log, and event values in this PDF were copied from the local NWIS synthetic PostgreSQL database. The document does not add wells, depths, causes, mitigations, or outcomes that are absent from those tables.